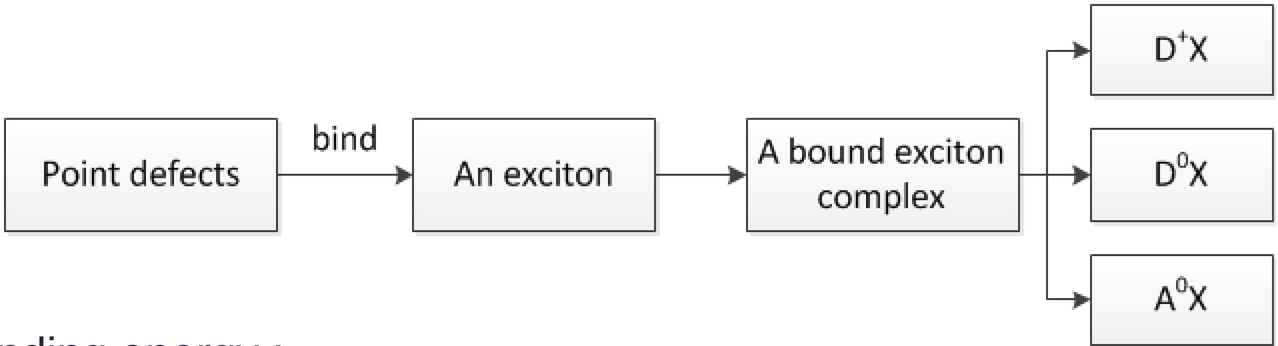
CHAPTER 14

Optical Properties of Bound
and Localized Excitons and of Defect States

Contents

- The optical properties of defect and localized states in bulk materials

14.1 Bound-Exciton and Multi-exciton Complexes



- The binding energy :

$$E^b_{D^+X} < E^b_{D^0X} < E^b_{A^0X}$$

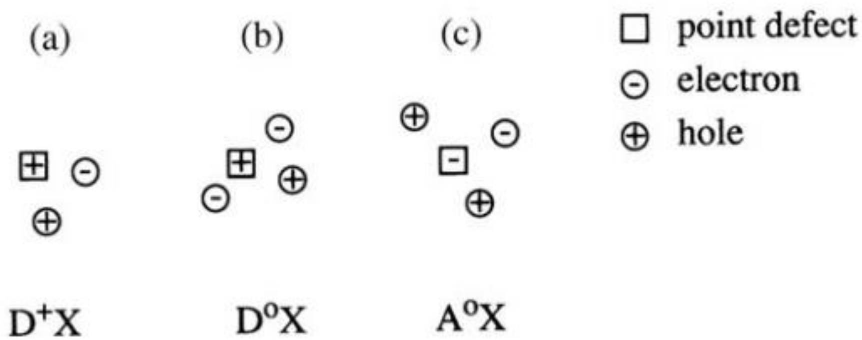
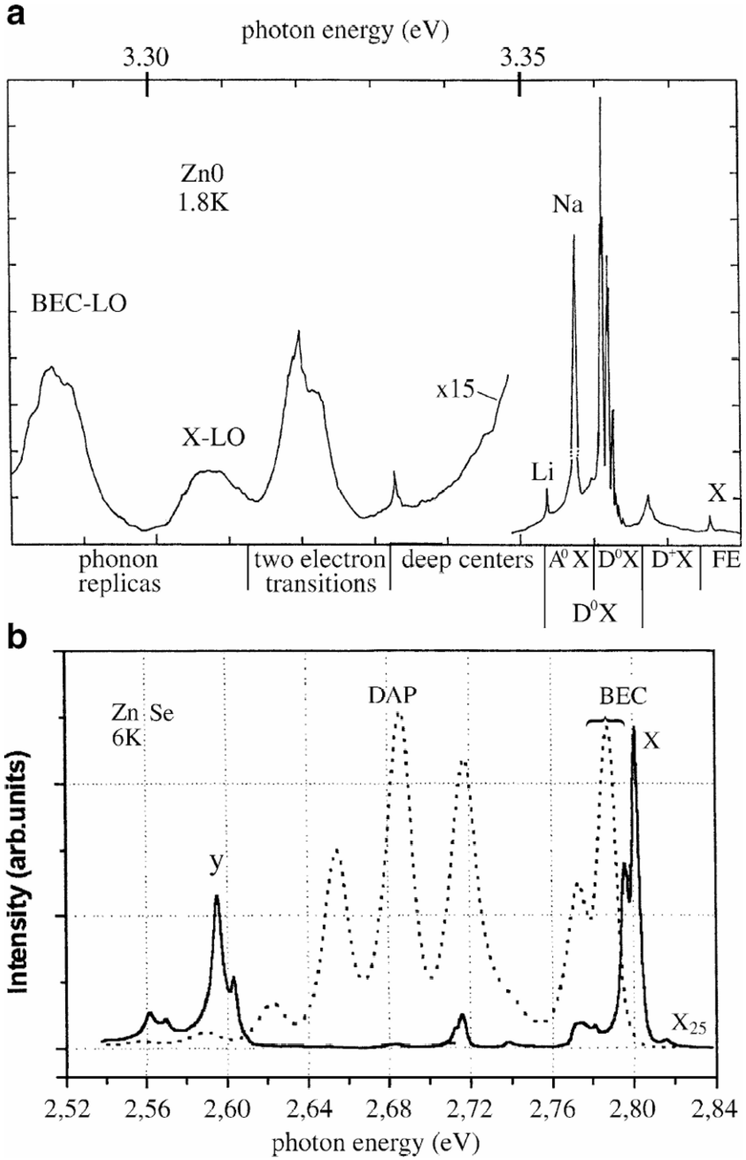


Fig. 14.1. Visualization of an exciton bound to an ionized donor (**a**), a neutral donor (**b**), and a neutral acceptor (**c**)

- For the D^0X complex :

$$E^b_{D^0X} \Big/ E^b_D = const \qquad 0.1\text{-}0.2\text{, Hayne's rule}$$

Fig. 14.2 Low-temperature and low-excitation luminescence spectra of ZnO (**a**) of two differently grown ZnSe samples (*solid line*: MBE growth with element sources, *dashed line*: with a compound source) (**b**) and of GaAs (**c**) [76H1, 76T1, 91V1, 03D1]



BEC often show up in luminescence and absorption spectra as extremely narrow peaks.

The low temperature luminescence of high quality samples is generally dominated by defect (or localiation-) related recombination processes

14.2 Photoluminescence of BEC in Indirect Semiconductor

- In some indirect semiconductors , multiple bound-exciton complexes can be formed,i.e., BEC which contain one, two, three or even more electron–hole pairs.
- BECs can be best observed at low lattice temperatures. With increasing T_L the BEC disappear depending on the material, often at 20~100 K.
- For optimal observation conditions: the point defects should be ≤10^{16}cm^{-3}, and BEC
  - are best observed in luminescence or photoluminescence excitation spectroscopy
  - only partly in absorption
  - only very rarely in reflection structures

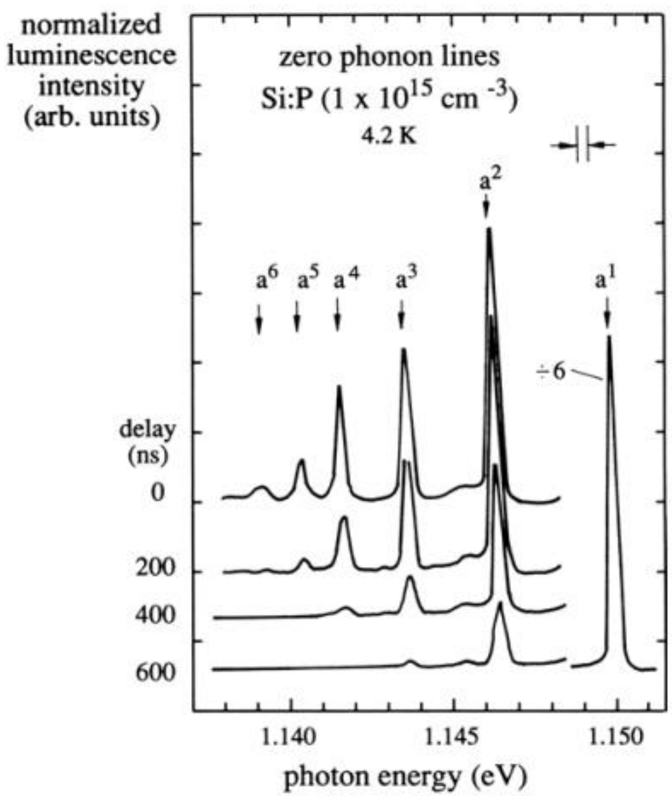


Fig. 14.8. The zero-phonon luminescence spectra of a multiple-bound-exciton complex in Si:P after pulsed excitation, normalized to the a_1 line [78T1]

14.3 Excitons in Disordered Systems

- Excitons can localized at band tail states
  - Whole excitons can be localized in potential fluctuations of sufficient depth and with diameters larger than their Bohr radius.
  - only one of the carriers is localized, usually the hole
- The absorption spectrum of disrodered system.
  - Region I: the absorption is weak, caused by impurities
  - Region II: the “Urbach tail” –localized states.
  - Region III: comprising the absorption via transitions into extended states.

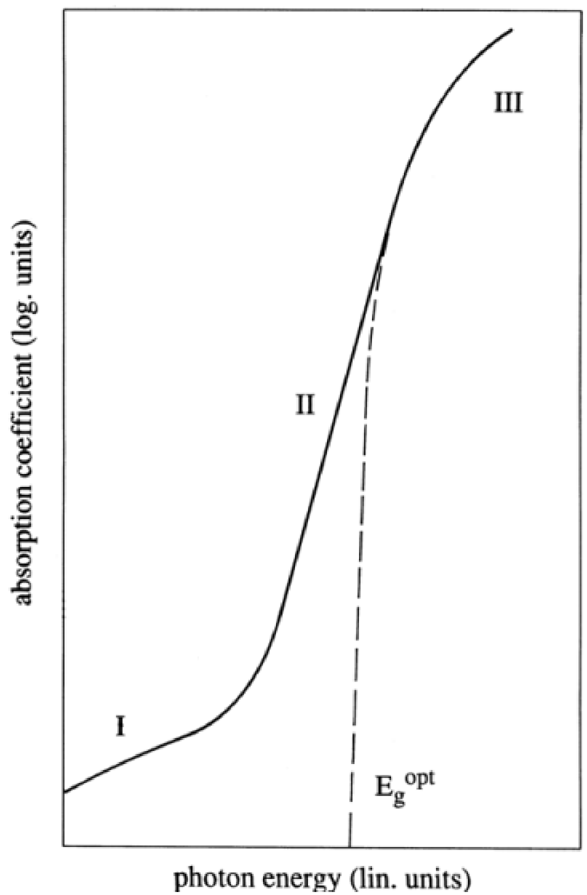


Fig. 14.12. The various regimes of the absorption coefficient in disordered semi-conductors [72W1]